ABDELRAHMAN ABDELREHIM

SENIOR MACHINE LEARNING ENGINEER / SPEECH AI / GENERATIVE AI

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PROFESSIONAL PROFILE

Machine Learning Engineer with approximately 5 years of experience developing applied AI and data science solutions for telecom Customer Care and Marketing. Focused on Arabic speech analytics, LLM systems, Whisper fine-tuning, RAG, and GPU inference optimization. Experienced in taking projects from data preparation and model evaluation to scalable workflows and stakeholder-facing insights.

PROFESSIONAL EXPERIENCE

Vodafone Egypt | Data Scientist

Feb 2023 – Oct 2026

Gen AI Arabic Speech Analytics Platform | Flagship project · Customer Care

- **End-to-end solution:** Built a large-scale Arabic call analytics workflow spanning audio preprocessing, ASR, LLM enrichment and TIBCO Spotfire business reporting.
- **Whisper fine-tuning:** Adapted Whisper to Egyptian Arabic customer-care calls using curated speech data and rigorous evaluation; reduced WER by approximately 50% relative to baseline.
- **Inference optimization:** Reduced transcription runtime by 87% with multiprocessing, dynamic batching and dual-GPU inference.
- **Intent classification:** Built hierarchical intent/sub-intent classification using dynamic few-shot prompting and LoRA fine-tuning of Qwen-14B, achieving up to 87% accuracy.
- **Summarization & privacy:** Developed LLM-driven call summaries, recurring-issue rollups and sensitive-data redaction to make conversations easier to review.
- **Follow-up promises:** Identified agent callback commitments and extracted promised times from transcripts to support checks on follow-up obligations.
- **Operational insights:** Combined sentiment, call KPIs and AI outputs into dashboards for business teams to analyze issues and service performance.

Conversational Voice Assistant | Standalone proof of concept

- Built an independent voice-assistant PoC using OpenAI APIs: speech-to-text, retrieval-augmented generation over knowledge content, LLM response generation and text-to-speech.

VODAFONE — MARKETING PROJECTS

Lifestyle Analyzer | Azure AI · Marketing

- Built a bilingual audio-to-insights workflow for customer interviews and lifestyle diaries using Azure speech and language models, extracting structured activities, places, brands and products.
- Mapped extracted locations and activities to customer segments to inform Marketing targeting and identify relevant interests for each segment.

Vodafone Maaky Campaign | KPI-based targeting · Marketing

- Analyzed customer KPIs to identify relevant audiences for Vodafone Maaky, an initiative connecting women who are not employed with work opportunities at Vodafone.

EARLIER EXPERIENCE

Upwork | Freelance Data Scientist — 2022 – 2023

- Delivered 14+ machine learning and data science projects to international clients; earned Top Rated status and a 100% Job Success Score.

Information Technology Institute (ITI) | AI Engineer Intern — Jan – Dec 2021

- Developed a real-time PPE detection application using YOLO, OpenCV, PyTorch and Streamlit.

TECHNICAL EXPERTISE

Programming & data: Python, SQL, PySpark, Pandas, NumPy, Apache Spark, Databricks

Speech & audio AI: Whisper, ASR fine-tuning, Azure AI Speech, VAD, audio preprocessing, WER evaluation

LLMs & GenAI: OpenAI APIs, Azure OpenAI, Qwen-14B, Hugging Face Transformers, LoRA / PEFT, RAG, embeddings, vector databases

ML & inference: PyTorch, TensorFlow, scikit-learn, model evaluation, CUDA, GPU optimization, dynamic batching

Analytics & reporting: Arabic NLP, customer segmentation, KPI-based targeting, TIBCO Spotfire, Power BI, Matplotlib

EDUCATION

B.Sc. Electronics & Communication Engineering | Alexandria University · 2018

AI-Pro Diploma | Information Technology Institute (ITI) · 2022

CERTIFICATIONS & TRAINING

AWS Certified Machine Learning – Specialty | FWD Advanced Data Analyst Program (ITIDA)